

animals. The same conclusion would have been reached under an Art. 53(a) EPC 1973 assessment. However, both tests were satisfied when the invention was restricted to mice.

b) Concept of "morality" and "ordre public"

The issue of morality and "ordre public" had been raised in T 356/93 (OJ 1995, 545), in connection with plants. The object of the invention was plants and seeds resistant to a particular class of herbicides so that they could be selectively protected against weeds and fungal diseases. This was achieved by stably integrating into the genome of the plants a heterologous DNA encoding a protein capable of inactivating or neutralising the herbicides. The patent was opposed under Art. 53(a) EPC 1973, in particular on the grounds that the exploitation of the invention was likely to cause serious damage to the environment.

The board held that the concept of **morality** was related to the belief that some behaviour was right and acceptable whereas other behaviour was wrong, this belief being founded on the totality of the accepted norms which were deeply rooted in a particular culture. For the purposes of the EPC, the culture in question was the culture inherent in European society and civilisation. Accordingly, inventions the exploitation of which was not in conformity with the conventionally accepted standards of conduct pertaining to this culture were to be excluded from patentability as being contrary to morality.

The board found that none of the claims related to subject-matter which could lead to a misuse or destructive use of plant biotechnological techniques because they concerned activities (production of plants and seeds, protection of plants from weeds or fungal diseases) and products (plant cells, plants, seeds) which could not be considered to be wrong as such in the light of the conventionally accepted standards of conduct of European culture. Plant biotechnology per se could not be regarded as being more contrary to public morality than traditional selective breeding.

In T 315/03 the board found that in the assessment of an Art. 53(a) EPC 1973 objection, no single definition of morality based on e.g. economic or religious principles represents an accepted standard in European culture. Opinion poll evidence was of very limited value for the reasons given in T 356/93 (OJ 1995, 545), where many drawbacks were identified, ranging from the type and the number of questions posed within one poll, through the size and representative nature of the cross-section of the population polled, to the manner of interpretation of the results obtained.

In T 356/93 the board defined the concept of **"ordre public"** as covering the protection of public security and the physical integrity of individuals as part of society. It also encompassed the protection of the environment. Accordingly, inventions the exploitation of which was likely to seriously prejudice the environment were to be excluded from patentability as being contrary to "ordre public". However, a decision in this respect presupposes that the threat to the environment be sufficiently substantiated at the time the decision is taken by the EPO.

In the specific case the board held that, although the documents submitted by the appellant (opponent) provided evidence of possible hazards from the application of genetic engineering techniques to plants, they did not lead to the definite conclusion that the exploitation of any of the claimed subject-matter would seriously prejudice the environment.

Regarding the compatibility with Art. 53(a) EPC 1973 of an invention involving genetically modified, herbicide-resistant plants, see also T.475/01.

In T.866/01 the invention related to euthanasia compositions which were used for producing humane death in "lower animals". The proprietor argued that the avowed use of the invention's teaching indicated in the patent ("bestimmungsgemäßer Gebrauch der erfindungsgemäßen Lehre") was the use of the claimed composition for mercy killing of lower animals and that this particular intended use did not infringe "ordre public" or morality. It was, in the proprietor's opinion not sufficient for an objection under Art. 100(a) and 53(a) EPC that the invention could also be exploited in a way that would infringe principles of morality or "ordre public". The board noted that no evidence had been provided that euthanasia of lower animals under the particular conditions described in the application would obviate any ethically based constitutional or other rules. The board therefore held that in the circumstances of the present case, Art. 53(a) EPC provided no legal basis for denying absolute substance protection for the claimed composition per se on the grounds that, with the exception of the intended use or exploitation, one or more of several or even all conceivable other exploitations or uses of the claimed composition (falling within the scope of protection) would be or could be regarded as breach of the principles of "ordre public" or morality, even if such conceivable exploitation might constitute a serious breach of the principle of "ordre public", such as a criminal offence including the killing of humans. The board noted that the mere possibility of abuse of the invention was not sufficient to deny patent protection pursuant to Art. 53(a) EPC.

c) Donor consent

The invention in T.1213/05 related to the human BRCA1 gene isolated from the genome, mutant forms of that gene and its use in the diagnosis of predisposition to breast and ovarian cancer. Appellant II submitted that the lack of proof that the donors of the cells had given previous informed consent to the commercial exploitation of the research results by patents as well as a benefit sharing agreement meant that the initial obtaining of these research results involved a violation of ordre public or morality as referred to in Art. 53(a) EPC.

The board disagreed. The EPC contained no provision establishing a requirement for applicants to submit evidence of previous informed consent of the donor of the cells or a benefit sharing agreement. That the legislator had not provided for a procedure of verifying such informed consent could also be deduced from the Biotech Directive, which had to be used as supplementary means of interpretation (R. 23b(1) EPC 1973, R. 26(1) EPC).

In addition, the board referred to ECJ judgment C-377/98 concerning the application for annulment of the Directive, where the Court had rejected the plea of the Kingdom of the

Netherlands that the absence of a provision requiring verification of the consent of the donor or recipient of products obtained by biotechnological means undermined the right to self-determination. The Directive concerned only the grant of patents and therefore did not extend to activities before and after grant, whether they involved research or the use of the patented products.

3. Patentability of biological inventions

3.1. Plants and plant varieties

According to Art. 53(b) EPC, a patent shall not be granted if the claimed subject-matter is directed to plant varieties. If a product claim does not specify any particular plant variety, but a **technical teaching which can be embodied in an unspecified number of plant varieties**, the subject-matter of the claimed invention does not relate to a plant variety or varieties within the meaning of Art. 53(b) EPC. Thus, a patent will not be granted for a single plant variety but can be granted if varieties may fall within the scope of the claims. If plant varieties are individually claimed, they are not patentable, irrespective of how they were made (G 1/98, OJ 2000, 111).

3.1.1 Definition of the term "plant varieties"

T 49/83 (OJ 1984, 112) first defined the term "plant varieties" as a multiplicity of plants which were largely the same in their characteristics and remained the same within specific tolerances after every propagation cycle. Following on from this, the board in T 320/87 (OJ 1990, 71) concluded that **hybrid seed and plants**, lacking stability in some trait of the whole generation population, could not be classified as plant varieties within the meaning of Art. 53(b) EPC 1973. In this regard, see also T 788/07. In T 356/93 (OJ 1995, 545) the board held that **plant cells** as such, which with modern technology can be cultured much like bacteria and yeasts, could not be considered to fall under the definition of a plant or of a plant variety. This was confirmed by G 1/98, which stated that plant cells should be treated like micro-organisms.

The term 'plant variety' is defined in R. 26(4) EPC in the same way as in Directive 98/44/EC on the legal protection of biotechnological inventions, namely as "any plant grouping within a single botanical taxon of the lowest known rank, which grouping, irrespective of whether the conditions for the grant of a plant variety are fully met, can be:

- (a) defined by the expression of the characteristics that results from a given genotype or combination of genotypes,
- (b) distinguished from any other plant grouping by the expression of at least one of the said characteristics, and
- (c) considered as a unit with regard to its suitability for being propagated unchanged."

The words "the expression of the characteristics that results from a given genotype or combination of genotypes" used in R. 26(4)(a) EPC refer to the entire constitution of a

plant or a set of genetic information. In contrast, a plant defined by single recombinant DNA sequences is not an individual plant grouping to which an entire constitution can be attributed within the above meaning. A plant defined in this way is not a concrete living being or grouping of concrete living beings but an abstract and open definition embracing an indefinite number of individual entities defined by a part of its genotype or by a property bestowed on it by that part (G 1/98, T 189/09, T 547/10).

In T 1208/12 the board held that a claim for a hybrid plant seed produced by a cross of two types of Brassica plants, was directed to an excluded plant variety. The board held that the technical situation underlying the present case was different from that in G 1/98 and that the claims were not for a seed or plant merely defined by the presence of a single recombinant DNA sequence. The definition of the claimed subject-matter thus did not fit the concept set out in G 1/98 of "an abstract and open definition embracing an indefinite number of individual entities defined by a part of its genotype or by a property bestowed on it by that part". In the case at hand, the subject-matter defined a seed or a plant which necessarily belongs to a particular plant grouping which complies with the definition of plant variety pursuant to R 26(4) EPC, i.e. it relates exclusively to individual plant groupings to which an entire constitution can be attributed.

3.1.2 Limits to patentability

In G 1/98, the Enlarged Board concluded that, where specific plant varieties were not claimed individually, the claim was not excluded from patentability under Art. 53(b) EPC, even if it might potentially embrace plant varieties. The Enlarged Board took the view that Art. 53(b) EPC 1973 defined the borderline between patent protection and plant variety protection. The extent of the exclusion for patents was the obverse of the availability of plant variety rights. Since plant variety rights were only granted for specific plant varieties and not for technical teachings which could be implemented in an indefinite number of plant varieties, it was not sufficient for the exclusion from patent protection in Art. 53(b) EPC 1973 to apply that one or more plant varieties were embraced or might be embraced by the claims of the patent application (see also T 475/01).

The Enlarged Board further held that Art. 64(2) EPC 1973 should not be taken into consideration when a claim to a process for the production of a plant variety was examined, in conformity with the established case law according to which the protection conferred by a process patent is extended to the **products obtained directly by the process**, even if the products are not patentable per se (see chapter II.A.7.1.).

Finally, the Enlarged Board held that the exception to patentability in Art. 53(b), first part of sentence, EPC 1973, applied to plant varieties irrespective of the way in which they were produced. Therefore, plant varieties containing genes introduced into an ancestral plant by recombinant gene technology were excluded from patentability. The underlying reason for this was that the exclusion in Art. 53(b) EPC 1973 was designed to exclude from patentability subject-matter which was eligible for protection under the plant breeders' rights system. It did not make any difference for the requirements under the UPOV Convention or under the Regulation on Plant Variety Rights, whether a variety was obtained by traditional breeding techniques or genetic engineering. The argument that

the EPC legislator had not envisaged the possibility of **genetically modified plant varieties** and therefore could not have intended to exclude them from patentability could not be accepted – laws were not restricted in their application to situations known to the legislator.

3.2. Animals and animal varieties

In T 19/90 (OJ 1990, 476) the board confirmed the general principle that the exception to patentability under Art. 53(b) EPC 1973 applies to certain categories of animals, but **not to animals as such**. In interpreting the term "animal varieties" the board emphasised the narrow interpretation to be given to the provisions of Art. 53(b) EPC 1973. It is not a bar to patentability for subject-matter which is not covered by any of the terms "animal varieties", "races animales" or "Tierarten".

Concerning the patentability of animals, it was held in T 315/03 (OJ 2006, 15) that, in an assessment under Art. 53(b) EPC 1973, the principle enunciated in G 1/98 (OJ 2000, 111) concerning plants and "plant varieties" should also be followed in the case of animals. This meant that a patent could not be granted for a single animal variety (or species or race, depending on which language text of the EPC 1973 was used) but could be granted if varieties might fall within the scope of its claims.

The bar on patenting under Art. 53(b), first part of sentence, EPC 1973 did not extend to the products of a micro-biological process, which were patentable under Art. 53(b), second part of sentence, EPC 1973. Thus patents were held to be grantable for **animals produced by a microbiological process**. However, particularly in cases of genetic manipulation of animals involving, as in the case at issue, the insertion of an activated oncogene, there are compelling reasons to consider the provisions of Art. 53(a) EPC when assessing patentability (T 19/90).

In T 315/03 the board did not agree with the assertion of some of the opponent appellants that the claimed transgenic mice were a new species because they inherited one particular characteristic, namely an increased probability of developing tumours. This was not enough to create a new species when the possible "starting material" might be any of a whole genus of animals, namely all mice. The board held that Art. 53(b) EPC 1973 did not exclude the patentability of the claims amended to encompass only mice and maintained the patent on that basis.

3.3. Essentially biological processes

Processes for the production of plants or animals are not patentable (under either the old or the revised version of the EPC) if they are essentially biological processes. Processes which are not essentially biological, on the other hand, are patentable.

3.3.1 Essentially biological processes for the production of animals

In T 19/90 (OJ 1990, 476) the board agreed that the process claims for the production of transgenic non-human mammals through chromosomal incorporation of an activated